

MAHMOUD ABDELNABY

SOC Analyst

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PROFESSIONAL SUMMARY

Computer & Systems Engineering graduate with 3+ years of hands-on security practice, targeting SOC Analyst and DFIR roles. Conducted structured forensic investigations, Windows event log/Sysmon analysis, PCAP-based network forensics, MITRE ATT&CK mapping, and timeline reconstruction across simulated APT scenarios. Proficient in Linux administration, Bash/PowerShell security automation, and AWS infrastructure, with penetration testing, malware analysis, and software engineering foundations.

TECHNICAL SKILLS

Security Operations & Digital Forensics: SOC Tier 1/2 · Alert Triage · Incident Investigation · SIEM (Splunk, ELK, Wazuh) · IOC Analysis · MITRE ATT&CK · Windows Forensics (Event Logs, Sysmon, Registry) · Network Forensics (Wireshark, tshark) · Email Forensics · Timeline Reconstruction · DFIR Reporting · Static Analysis · PE Analysis · x86/x64 Assembly Analysis · Ghidra · WinDbg · GDB

Penetration Testing: Network Enumeration · Web Security Testing · Nmap · Burp Suite · Metasploit · Hydra · Hashcat · SQLmap

Systems & Networking: Linux (RHEL, Ubuntu, Kali) · ssh · TCP/IP · DNS · DHCP · VLANs · Routing & Switching · NAT · ACLs · systemd · journalctl

Cloud & DevOps: AWS (EC2, S3, Lambda, VPC, IAM, API Gateway, CloudFront) · Docker · Git · Bash · PowerShell · Terraform · Ansible

Programming: Python · Bash · PowerShell · C++ · Java · JavaScript · TypeScript · Assembly (x86/x64)

PROJECTS & TECHNICAL EXPERIENCE

Digital Forensics & Incident Response Investigations

Hands-on Security Labs — [Reports | Mahmoud Abdelnaby](#)

Conducted end-to-end DFIR investigations on simulated incidents, reconstructing attack chains and documenting findings per structured IR methodology.

- Investigated an APT spear-phishing intrusion — analyzed email headers, PCAPs, and PowerShell Script Block logs (EventID 4104) tracing PowerShell C2, credential theft via Sticky Notes/KeePass, and DNS exfiltration
- Reconstructed a CVE-2022-30190 (Follina) compromise chain via Sysmon telemetry (Event IDs 1, 3, 11, 22) — payload delivery, privilege escalation to SYSTEM, and persistence mechanisms
- Mapped adversary behavior to MITRE ATT&CK (20+ techniques), extracted IOCs (domains, IPs, hashes), and built minute-level timelines using Timeline Explorer and SysmonView
- Compiled a DFIR methodology reference covering evidence acquisition, volatile/non-volatile analysis, and forensic reporting per NIST SP 800-86 and SP 800-61

Tools: Wireshark · tshark · Sysmon · SysmonView · Timeline Explorer · Inkpase · VirusTotal

Hands-on SOC & Security Labs on tryhackme and shiftzero labs | [tryhackme profile](#)

Practiced SOC operations and incident investigation through TryHackMe (SOC Level 1 & 2, Top 2%) and ShiftZero SOC labs. Performed alert triage and security event analysis across SIEM platforms (Splunk, ELK, Wazuh), investigating Windows Event Logs, Sysmon telemetry, and network captures to classify alerts, extract IOCs, and correlate indicators across log sources. Practiced endpoint investigation, PowerShell execution analysis, email

header forensics, authentication event review, and network traffic inspection with Wireshark. Applied MITRE ATT&CK mapping, structured investigation workflows, and escalation reasoning across simulated security incidents.

Infrastructure & Security Monitoring Automation

Systems Automation & Cloud Engineering | 2/2026

- Developed a Bash-based Linux security monitoring script automating failed SSH auth detection, sudo escalation auditing, sudoers enumeration, service monitoring (sshd, docker, ufw), log analysis via journalctl, and network health checks
- Automated AWS infrastructure provisioning — VPC with public/private subnets, NACLs, route tables, and internet gateway via idempotent Bash/AWS CLI scripts
- Built a serverless deployment script provisioning DynamoDB, Lambda, API Gateway, S3, IAM (least-privilege policies), and ACM TLS certificates

Tools: Bash · AWS CLI · journalctl · systemd · ss · mpstat · sar

Qanoony — Legal Services Platform

Graduation Project — Zagazig University | 7/2026

Legal services platform connecting clients with lawyers. Contributed backend modules for Search, Services, Payments, and Consultations.

- Payments: Integrated Stripe Checkout with webhook signature verification; enforced ownership authorization, duplicate payment prevention, and automated refund handling
- Consultations: Developed booking management logic with role-based access control (Client/Lawyer/Admin) and state transition validation
- Services: Built service catalog APIs with Zod schema validation for data integrity and payment-linked request confirmation
- Search: Implemented cross-entity search API with regex-escaped input sanitization and role-scoped result filtering

Stack: Node.js · TypeScript · Express.js · MongoDB · Stripe · Zod

SnapLoad — Desktop Video Downloader

[Mabdelnaby1924/SnapLoad: A modern desktop GUI for yt-dlp](#) | 3/2026

Identified a use case and designed a desktop GUI wrapping yt-dlp. Defined requirements, workflow, and architecture; used AI-assisted development for implementation.

- Designed application architecture with separated core/UI layers, threaded downloads, real-time progress tracking, and error parsing
- Built and packaged as a standalone Windows executable with Python/CustomTkinter

Stack: Python · CustomTkinter · yt-dlp · PyInstaller

EDUCATION

Bachelor of Science (B.Sc.) in Computer & Systems Engineering

Zagazig University — Faculty of Engineering | 2026

CERTIFICATIONS & COURSES

CCNA

RHCSA — Self-Study (*Red Hat System Administration*)

SOC Level 1 & Level 2 — (*TryHackMe*)

Jr Penetration Tester — (*TryHackMe*)

OSCP — Self-Study

AWS Cloud Practitioner — Self-Study (*AWS Foundational*)